

Qualification Pack



Soil Pollution Monitoring Specialist

QP Code: SGJ/Q6901

Version: 1.0

NSQF Level: 5.5

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi - 110021 || email:sarvesh@sscgj.in

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SGJ/Q6901: Soil Pollution Monitoring Specialist

Brief Job Description

The Soil Pollution Monitoring Specialist monitors and manages soil health by analyzing composition, quality, and pollution levels. The role includes soil classification, sampling, and data analysis using statistical and GIS tools. Specialists also develop pollution mitigation strategies and promote advanced technologies for sustainable soil and environmental management.

Personal Attributes

Analytical, detail-oriented, adaptable, environmentally conscious, collaborative, and tech-savvy.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N6901: Introduce Soil Composition, Quality, and Health Implications](#)
2. [SGJ/N6902: Assess and Classify Soil Properties](#)
3. [SGJ/N6903: Identify techniques for Soil Sampling and Ensuring Quality Standards](#)
4. [SGJ/N6904: Apply Statistical and GIS Tools in Soil Analysis](#)
5. [SGJ/N6905: Prepare Strategies for Soil Pollution Mitigation and Adoption of Advanced Technologies](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Sustainability
Occupation	Soil Scientist
Country	India
NSQF Level	5.5
Credits	20
Aligned to NCO/ISCO/ISIC Code	2132.0300 (NCO, 2015), Soil Scientist

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Minimum Educational Qualification & Experience	Completed 3 year UG degree (Environmental Science/Agriculture/soil science/Environmental Engineer/Chemistry) with NA of experience OR Completed 2nd year of UG (UG Diploma) (Environmental Science/Agriculture/soil science/Environmental Engineer/Chemistry) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (4.5) with 3 Years of experience Environmental science/ NABL Accredited- Testing Labs/ NABET Accredited- Labs OR Previous relevant Qualification of NSQF Level (5) with 1.5 years of experience Environmental science
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-5.5-ES-04263-2025-V1
NQR Version	1

Remarks:

Total 600 Hours: (Theory: 200 hours with employability skills + Practical: 190 hours + 210 hours of On-the-Job Training (OJT))

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SGJ/N6901: Introduce Soil Composition, Quality, and Health Implications

Description

It provides essential knowledge and tools to evaluate soil health, identify contaminants, assess ecological and human health risks, and implement effective soil conservation strategies.

Scope

The scope covers the following :

- This focuses on the Earth lithosphere, soil formation, classification, pollution, risk assessment, and sustainable soil management.

Elements and Performance Criteria

Introduction to Soil, Pollution, Quality and relation to human.

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the structure and composition of the Earth's lithosphere.
- PC2.** Explain the processes involved in forming and transforming rocks and minerals
- PC3.** Identify different types of rocks and their properties.
- PC4.** Describe the factors influencing soil formation (parent material, climate, organisms, topography, time).
- PC5.** Explain the processes of soil development (weathering, leaching, accumulation, etc.).
- PC6.** Identify various sources of soil pollution (industrial, agricultural, urban) and recognize different types of pollutants (heavy metals, pesticides, organic contaminants).
- PC7.** Understand the adverse effects of soil pollution on soil health, ecosystem function, food safety, and human health.
- PC8.** Describe the processes governing the movement and transformation of pollutants in soil, including adsorption, leaching, and biodegradation
- PC9.** Identify different soil types and their characteristics.
- PC10.** Identify different types of soil pollutants (heavy metals, organic contaminants, pesticides, etc.) and their sources.
- PC11.** Describe the key components of an ecological risk assessment (problem formulation, exposure assessment, effects assessment, risk characterization).
- PC12.** Evaluate the potential adverse effects of soil pollution on living organisms.
- PC13.** Characterize the ecological risks and develop risk management strategies.
- PC14.** Define soil quality and identify key physical, chemical, and biological indicators used to assess soil health and function.
- PC15.** Understand the causes and consequences of soil degradation, including erosion, compaction, nutrient depletion, and salinization.
- PC16.** Learn methods for monitoring soil quality, interpreting data, and making informed management decisions.

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- PC17.** Identify common sources of soil contamination (industrial, agricultural, waste disposal) and recognize different types of contaminants (heavy metals, pesticides, organic pollutants).
- PC18.** Understand how contaminants move through the environment and enter the human body via inhalation, ingestion, and dermal contact.
- PC19.** Explore the acute and chronic health effects associated with exposure to various soil contaminants.
- PC20.** Learn about methods for assessing and managing risks posed by soil contamination to human health.
- PC21.** Learn proper techniques for soil identification.
- PC22.** Identify different soil types in the field based on their physical characteristics.
- PC23.** Interpret soil type results and relate them to soil quality and contamination.
- PC24.** Learn about techniques for soil erosion control, nutrient management, and soil health improvement.
- PC25.** Compute risk assessment indices to evaluate the ecological risks associated with soil pollution of different natures.
- PC26.** Identify management strategies to mitigate the identified risks.
- PC27.** Interpret soil test results and evaluate soil quality using various assessment tools and indices.
- PC28.** Determine potential health risks to human populations under exposure to different soil contaminants.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the structure and composition of the Earth's lithosphere and the processes of rock and mineral formation.
- KU2.** Gain knowledge of soil formation factors, classification, and development processes such as weathering and leaching.
- KU3.** Identify sources and types of soil pollutants and understand their impact on soil health, ecosystems, and human well-being.
- KU4.** Comprehend the transport, transformation, and fate of soil pollutants through processes like adsorption, leaching, and biodegradation.
- KU5.** Understand ecological and human health risk assessment frameworks and their application in evaluating soil contamination.
- KU6.** Recognize indicators of soil quality, threats of degradation, and effective soil conservation, monitoring, and management practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to observe, classify, and analyze different types of rocks, soils, and their properties in field and lab settings.

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- GS2.** Skill in identifying soil pollutants and interpreting their sources, behavior, and impact on ecosystems and human health.
- GS3.** Competence in conducting soil quality assessments using scientific tools, test results, and relevant indicators.
- GS4.** Proficiency in applying ecological and human health risk assessment frameworks for contaminated soil sites.
- GS5.** Capability to recommend and implement soil conservation, pollution control, and sustainable land management practices.
- GS6.** Effective communication and reporting skills to present findings and risk mitigation strategies to stakeholders.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Soil, Pollution, Quality and relation to human.</i>	64	36	-	-
PC1. Describe the structure and composition of the Earth's lithosphere.	2	-	-	-
PC2. Explain the processes involved in forming and transforming rocks and minerals	1	-	-	-
PC3. Identify different types of rocks and their properties.	2	-	-	-
PC4. Describe the factors influencing soil formation (parent material, climate, organisms, topography, time).	8	-	-	-
PC5. Explain the processes of soil development (weathering, leaching, accumulation, etc.).	1	-	-	-
PC6. Identify various sources of soil pollution (industrial, agricultural, urban) and recognize different types of pollutants (heavy metals, pesticides, organic contaminants).	2	-	-	-
PC7. Understand the adverse effects of soil pollution on soil health, ecosystem function, food safety, and human health.	4	-	-	-
PC8. Describe the processes governing the movement and transformation of pollutants in soil, including adsorption, leaching, and biodegradation	2	-	-	-
PC9. Identify different soil types and their characteristics.	4	-	-	-
PC10. Identify different types of soil pollutants (heavy metals, organic contaminants, pesticides, etc.) and their sources.	6	-	-	-
PC11. Describe the key components of an ecological risk assessment (problem formulation, exposure assessment, effects assessment, risk characterization).	1	-	-	-
PC12. Evaluate the potential adverse effects of soil pollution on living organisms.	4	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Characterize the ecological risks and develop risk management strategies.	5	-	-	-
PC14. Define soil quality and identify key physical, chemical, and biological indicators used to assess soil health and function.	2	-	-	-
PC15. Understand the causes and consequences of soil degradation, including erosion, compaction, nutrient depletion, and salinization.	3	-	-	-
PC16. Learn methods for monitoring soil quality, interpreting data, and making informed management decisions.	2	-	-	-
PC17. Identify common sources of soil contamination (industrial, agricultural, waste disposal) and recognize different types of contaminants (heavy metals, pesticides, organic pollutants).	4	-	-	-
PC18. Understand how contaminants move through the environment and enter the human body via inhalation, ingestion, and dermal contact.	2	-	-	-
PC19. Explore the acute and chronic health effects associated with exposure to various soil contaminants.	4	-	-	-
PC20. Learn about methods for assessing and managing risks posed by soil contamination to human health.	5	-	-	-
PC21. Learn proper techniques for soil identification.	-	8	-	-
PC22. Identify different soil types in the field based on their physical characteristics.	-	2	-	-
PC23. Interpret soil type results and relate them to soil quality and contamination.	-	6	-	-
PC24. Learn about techniques for soil erosion control, nutrient management, and soil health improvement.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. Compute risk assessment indices to evaluate the ecological risks associated with soil pollution of different natures.	-	4	-	-
PC26. Identify management strategies to mitigate the identified risks.	-	8	-	-
PC27. Interpret soil test results and evaluate soil quality using various assessment tools and indices.	-	4	-	-
PC28. Determine potential health risks to human populations under exposure to different soil contaminants.	-	1	-	-
NOS Total	64	36	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N6901
NOS Name	Introduce Soil Composition, Quality, and Health Implications
Sector	Green Jobs
Sub-Sector	
Occupation	Soil Scientist
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N6902: Assess and Classify Soil Properties

Description

This integrates soil texture, structure, nutrients, organic matter, microbial activity, and advanced analytical techniques to evaluate soil quality, support crop productivity, and guide sustainable land management.

Scope

The scope covers the following :

- To comprehensively study soil's physical, chemical, and biological properties, processes, and analysis methods to assess soil health, fertility, and its role in sustainable plant growth and environmental management.

Elements and Performance Criteria

Soil Properties

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the concept of soil texture, its classification based on particle size distribution (sand, silt, clay), and its influence on other soil properties and functions.
- PC2.** Learn about soil structure, the formation and types of soil aggregates, their role in soil porosity, water infiltration, and root growth.
- PC3.** Grasp the concept of soil water holding capacity, factors affecting it (texture, structure, organic matter), and its importance for plant growth and water management.
- PC4.** Explore additional physical properties such as soil bulk density, particle density, porosity, soil temperature, and colour, and their implications for soil management and plant growth.
- PC5.** Understand the role of macro (N, P, K, Ca, Mg, S) & micronutrients (Fe, Mn, Zn, Cu, B, Mo, Cl) in plant growth and soil health.
- PC6.** Learn about the composition, decomposition, and functions of soil organic matter and its significance as a carbon sink.
- PC7.** Grasp the concepts of soil pH, salinity, and electrical conductivity and their influence on nutrient availability, plant growth, and soil health.
- PC8.** Identify common heavy metals and toxic ions in soil, their sources, and their potential adverse effects on ecosystems and human health.
- PC9.** Understand the diversity of soil organisms (bacteria, fungi, etc.) and their crucial roles in nutrient cycling, decomposition, and soil structure formation.
- PC10.** Learn about the types and functions of soil enzymes (oxidoreductases, hydrolases, transferases, etc.) and their significance as indicators of soil biological activity and health.
- PC11.** Metagenomics and Meta transcriptomics: Understand the principles and applications of metagenomics and meta transcriptomics data in studying soil microbial communities and their functional potential
- PC12.** Conduct laboratory tests (e.g., hydrometer method, feel method) to determine soil texture and classify soil samples.

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- PC13.** Assess soil aggregate stability using wet sieving or other methods and relate it to soil structure and management practices.
- PC14.** Determine soil water holding capacity using field or laboratory methods and interpret the results for irrigation and drainage planning
- PC15.** Measure soil bulk density and calculate porosity to evaluate soil compaction and its impact on root growth and water movement.
- PC16.** Conduct laboratory tests to determine levels of macro and micronutrients in soil samples.
- PC17.** Analyze soil organic carbon and estimate soil organic matter content using standard methods
- PC18.** Measure soil pH, salinity, and electrical conductivity using appropriate instruments and techniques.
- PC19.** Employ analytical techniques (e.g., Inductively coupled plasma) to detect and quantify heavy metals and toxic ions using Ion chromatography in soil.
- PC20.** Conduct laboratory assays to measure the activity of key soil enzymes, such as dehydrogenase, urease, and phosphatase.
- PC21.** Analyze soil DNA and RNA sequences to identify microbial communities and their functional genes and transcripts.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand soil physical properties (texture, structure, porosity, water holding capacity) and their impact on plant growth and soil function.
- KU2.** Gain knowledge of essential macro- and micronutrients, their roles in soil fertility, and plant nutrition.
- KU3.** Comprehend the significance of organic matter and soil carbon in enhancing soil health and acting as a carbon sink.
- KU4.** Understand chemical properties like pH, salinity, and electrical conductivity and their influence on nutrient availability and plant performance.
- KU5.** Recognize the role of soil biology, including microorganisms and enzymes, in nutrient cycling, decomposition, and ecosystem functioning.
- KU6.** Acquire skills in laboratory and field techniques for analyzing soil properties, nutrient content, contaminants, and microbial community structure.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to perform precise soil sampling, laboratory testing, and field measurements for physical, chemical, and biological soil analysis.
- GS2.** Proficiency in using scientific instruments and techniques for determining soil texture, nutrients, pH, salinity, and contaminant levels.
- GS3.** Competence in interpreting soil data to assess fertility, water retention, microbial activity, and overall soil health.



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- GS4.** Skills in applying metagenomic and metatranscriptomic tools for analyzing soil microbial communities and their functions.
- GS5.** Capability to link soil properties with crop requirements and recommend suitable soil and water management practices.
- GS6.** Effective communication and documentation skills to report findings, collaborate with stakeholders, and support sustainable agriculture decisions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Soil Properties</i>	45	55	-	-
PC1. Understand the concept of soil texture, its classification based on particle size distribution (sand, silt, clay), and its influence on other soil properties and functions.	4	-	-	-
PC2. Learn about soil structure, the formation and types of soil aggregates, their role in soil porosity, water infiltration, and root growth.	7	-	-	-
PC3. Grasp the concept of soil water holding capacity, factors affecting it (texture, structure, organic matter), and its importance for plant growth and water management.	2	-	-	-
PC4. Explore additional physical properties such as soil bulk density, particle density, porosity, soil temperature, and colour, and their implications for soil management and plant growth.	5	-	-	-
PC5. Understand the role of macro (N, P, K, Ca, Mg, S) & micronutrients (Fe, Mn, Zn, Cu, B, Mo, Cl) in plant growth and soil health.	3	-	-	-
PC6. Learn about the composition, decomposition, and functions of soil organic matter and its significance as a carbon sink.	2	-	-	-
PC7. Grasp the concepts of soil pH, salinity, and electrical conductivity and their influence on nutrient availability, plant growth, and soil health.	2	-	-	-
PC8. Identify common heavy metals and toxic ions in soil, their sources, and their potential adverse effects on ecosystems and human health.	7	-	-	-
PC9. Understand the diversity of soil organisms (bacteria, fungi, etc.) and their crucial roles in nutrient cycling, decomposition, and soil structure formation.	7	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Learn about the types and functions of soil enzymes (oxidoreductases, hydrolases, transferases, etc.) and their significance as indicators of soil biological activity and health.	2	-	-	-
PC11. Metagenomics and Meta transcriptomics: Understand the principles and applications of metagenomics and meta transcriptomics data in studying soil microbial communities and their functional potential	4	-	-	-
PC12. Conduct laboratory tests (e.g., hydrometer method, feel method) to determine soil texture and classify soil samples.	-	2	-	-
PC13. Assess soil aggregate stability using wet sieving or other methods and relate it to soil structure and management practices.	-	5	-	-
PC14. Determine soil water holding capacity using field or laboratory methods and interpret the results for irrigation and drainage planning	-	8	-	-
PC15. Measure soil bulk density and calculate porosity to evaluate soil compaction and its impact on root growth and water movement.	-	3	-	-
PC16. Conduct laboratory tests to determine levels of macro and micronutrients in soil samples.	-	7	-	-
PC17. Analyze soil organic carbon and estimate soil organic matter content using standard methods	-	7	-	-
PC18. Measure soil pH, salinity, and electrical conductivity using appropriate instruments and techniques.	-	8	-	-
PC19. Employ analytical techniques (e.g., Inductively coupled plasma) to detect and quantify heavy metals and toxic ions using Ion chromatography in soil.	-	4	-	-
PC20. Conduct laboratory assays to measure the activity of key soil enzymes, such as dehydrogenase, urease, and phosphatase.	-	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Analyze soil DNA and RNA sequences to identify microbial communities and their functional genes and transcripts.	-	5	-	-
NOS Total	45	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N6902
NOS Name	Assess and Classify Soil Properties
Sector	Green Jobs
Sub-Sector	
Occupation	Soil Scientist
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N6903: Identify techniques for Soil Sampling and Ensuring Quality Standards

Description

This equips individuals with expertise in various soil sampling techniques, grid-based sampling, preservation protocols, microbial community assessment, and analysis of soil health indicators under varying environmental conditions.

Scope

The scope covers the following :

- This covers the knowledge and practical skills required for soil sampling, handling, grid-based study design, microbial analysis, and quality assessment for evaluating soil health and ecosystem functions.

Elements and Performance Criteria

Soil Sampling, Procedure and Quality Assessment

To be competent, the user/individual on the job must be able to:

- PC1.** Learn about different soil sampling methods (e.g., auger sampling, core sampling, grab sampling, composite sampling) and their suitability for different soil types for specific purposes.
- PC2.** Grasp the concept of grid-based sampling, its advantages in ensuring representative sampling, and its application in spatial data analysis along with Grid-based Study and Geo-tagging
- PC3.** Learn proper techniques for handling, labelling, and preserving soil samples to maintain their integrity and prevent any contamination.
- PC4.** Understand the diversity and functions of soil microbial communities and their role in maintaining soil health and ecosystem services.
- PC5.** Identify key microbial and chemical markers (e.g., microbial biomass, enzyme activities, community structure, nutrients, etc) that can be used to assess soil quality and function.
- PC6.** Explore how microbial communities respond to various stressors (e.g., pollution, land-use change, climate change) and their implications for soil health assessment.
- PC7.** Conduct hands-on soil sampling using various techniques in different field conditions.
- PC8.** Design and implement a grid-based sampling plan for a specific area, considering factors like variability, objectives, and resources.
- PC9.** Prepare soil samples for laboratory analysis and interpret the results in relation to sampling objectives.
- PC10.** Microbial Biomass and Activity Measurements: Conduct laboratory assays to determine microbial biomass and activity in soil samples.
- PC11.** Molecular Analysis of Soil Microbial Communities: Extract DNA from soil samples and perform basic molecular techniques (e.g., PCR, gel electrophoresis, RAPD, Biolog, etc.) for microbial community analysis.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Types and applications of soil sampling methods.
- KU2.** Principles and benefits of grid-based sampling and geo-tagging.
- KU3.** Techniques for proper sample handling, labelling, and preservation.
- KU4.** Role of soil microbes in soil health and ecosystem function.
- KU5.** Key indicators for assessing soil quality and microbial activity.
- KU6.** Basic lab techniques for microbial and molecular analysis.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to perform accurate soil sampling using appropriate techniques in diverse field conditions.
- GS2.** Skill in designing and executing grid-based sampling plans with geo-tagging.
- GS3.** Competence in handling, labelling, and preserving soil samples to avoid contamination.
- GS4.** Proficiency in preparing samples and conducting lab assays for microbial and chemical analysis.
- GS5.** Basic molecular biology skills including DNA extraction, PCR, and gel electrophoresis.
- GS6.** Analytical skills to interpret soil data for quality assessment and environmental impact.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Soil Sampling, Procedure and Quality Assessment</i>	58	42	-	-
PC1. Learn about different soil sampling methods (e.g., auger sampling, core sampling, grab sampling, composite sampling) and their suitability for different soil types for specific purposes.	8	-	-	-
PC2. Grasp the concept of grid-based sampling, its advantages in ensuring representative sampling, and its application in spatial data analysis along with Grid-based Study and Geo-tagging	10	-	-	-
PC3. Learn proper techniques for handling, labelling, and preserving soil samples to maintain their integrity and prevent any contamination.	10	-	-	-
PC4. Understand the diversity and functions of soil microbial communities and their role in maintaining soil health and ecosystem services.	10	-	-	-
PC5. Identify key microbial and chemical markers (e.g., microbial biomass, enzyme activities, community structure, nutrients, etc) that can be used to assess soil quality and function.	10	-	-	-
PC6. Explore how microbial communities respond to various stressors (e.g., pollution, land-use change, climate change) and their implications for soil health assessment.	10	-	-	-
PC7. Conduct hands-on soil sampling using various techniques in different field conditions.	-	8	-	-
PC8. Design and implement a grid-based sampling plan for a specific area, considering factors like variability, objectives, and resources.	-	9	-	-
PC9. Prepare soil samples for laboratory analysis and interpret the results in relation to sampling objectives.	-	7	-	-
PC10. Microbial Biomass and Activity Measurements: Conduct laboratory assays to determine microbial biomass and activity in soil samples.	-	10	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Molecular Analysis of Soil Microbial Communities: Extract DNA from soil samples and perform basic molecular techniques (e.g., PCR, gel electrophoresis, RAPD, Biolog, etc.) for microbial community analysis.	-	8	-	-
NOS Total	58	42	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N6903
NOS Name	Identify techniques for Soil Sampling and Ensuring Quality Standards
Sector	Green Jobs
Sub-Sector	
Occupation	Soil Scientist
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N6904: Apply Statistical and GIS Tools in Soil Analysis

Description

This covers the use of statistical analysis, hypothesis testing, and geospatial techniques to assess soil data, identify patterns, and create visual outputs using tools like SPSS, Excel, Q-GIS, and XLSTAT.

Scope

The scope covers the following :

- To analyze, interpret, and visualize soil data using statistical methods and geospatial tools for informed decision-making and effective presentation.

Elements and Performance Criteria

Soil Analysis and significance of Statistical and RS-GIS Tools

To be competent, the user/individual on the job must be able to:

- PC1.** Understand and apply measures of central tendency (mean, median, mode), dispersion (range, variance, standard deviation), and distribution to summarize and describe soil data
- PC2.** Learn about hypothesis testing, confidence intervals, and statistical significance to draw inferences and make generalizations from soil data.
- PC3.** Explore the relationships between different soil variables using correlation and regression analysis, and interpret the results to understand their interdependence.
- PC4.** Introduce geospatial concepts based on RS-GIS tools for analyzing and mapping geo-referenced soil data.
- PC5.** Perform descriptive and inferential statistical analyses using SPSS or Excel, including t-tests, ANOVA, and regression analysis.
- PC6.** Employ Q-GIS or similar software to create maps, visualize spatial patterns, and perform geostatistical analysis on soil data.
- PC7.** Use XLSTAT or other tools to create informative charts, graphs, and maps to present soil data and analysis results effectively.
- PC8.** Using Q-GIS to view raster and vector data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic statistical concepts such as mean, median, mode, range, variance, and standard deviation for soil data analysis.
- KU2.** Principles of hypothesis testing, confidence intervals, and statistical significance for drawing valid inferences from soil datasets.
- KU3.** Understanding of correlation and regression techniques to examine relationships among soil variables.
- KU4.** Fundamentals of geospatial analysis and RS-GIS tools for handling and interpreting geo-referenced soil data.

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- KU5.** Proficiency in using statistical software like SPSS, Excel, and XLSTAT for performing descriptive and inferential analyses.
- KU6.** Knowledge of Q-GIS for viewing raster/vector data, mapping, and conducting geostatistical analysis of soil parameters.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyze and interpret soil data using statistical tools and methods.
- GS2.** Proficiency in using SPSS, Excel, and XLSTAT for performing statistical and graphical analysis.
- GS3.** Skill in conducting correlation, regression, t-tests, and ANOVA for soil data interpretation.
- GS4.** Competence in using Q-GIS and RS-GIS tools for mapping, spatial analysis, and visualizing soil patterns.
- GS5.** Capability to handle and interpret raster and vector data within geospatial platforms.
- GS6.** Effective presentation of soil data insights through charts, graphs, and thematic maps.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Soil Analysis and significance of Statistical and RS-GIS Tools</i>	60	40	-	-
PC1. Understand and apply measures of central tendency (mean, median, mode), dispersion (range, variance, standard deviation), and distribution to summarize and describe soil data	10	-	-	-
PC2. Learn about hypothesis testing, confidence intervals, and statistical significance to draw inferences and make generalizations from soil data.	10	-	-	-
PC3. Explore the relationships between different soil variables using correlation and regression analysis, and interpret the results to understand their interdependence.	20	-	-	-
PC4. Introduce geospatial concepts based on RS-GIS tools for analyzing and mapping geo-referenced soil data.	20	-	-	-
PC5. Perform descriptive and inferential statistical analyses using SPSS or Excel, including t-tests, ANOVA, and regression analysis.	-	10	-	-
PC6. Employ Q-GIS or similar software to create maps, visualize spatial patterns, and perform geostatistical analysis on soil data.	-	10	-	-
PC7. Use XLSTAT or other tools to create informative charts, graphs, and maps to present soil data and analysis results effectively.	-	10	-	-
PC8. Using Q-GIS to view raster and vector data.	-	10	-	-
NOS Total	60	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N6904
NOS Name	Apply Statistical and GIS Tools in Soil Analysis
Sector	Green Jobs
Sub-Sector	
Occupation	Soil Scientist
NSQF Level	5.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N6905: Prepare Strategies for Soil Pollution Mitigation and Adoption of Advanced Technologies

Description

The covers bioremediation using plants and microbes, along with IPR fundamentals and practical skills in patent analysis, field evaluation, and case-based learning.

Scope

The scope covers the following :

- The explores bioremediation with plants and microbes in saline soils and introduces IPR essentials, supported by fieldwork, case studies, and patent analysis.

Elements and Performance Criteria

Soil Pollution: Mitigation and Technologies

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the principles and mechanisms of bioremediation, including the use of microorganisms and plants to degrade or remove pollutants from soil.
- PC2.** Explore the use of plants for the remediation of contaminated soil, including phytoextraction, Phyto stabilization, phytodegradation, and rhizofiltration.
- PC3.** Learn about the use of salt-tolerant plants and microorganisms to improve soil quality and productivity in saline environments.
- PC4.** Understand the role of microorganisms in bioremediation and the factors influencing their effectiveness in degrading various pollutants.
- PC5.** Understand the concept and types of IPR, including patents, trademarks, copyrights, and trade secrets, and their importance in protecting innovations and promoting economic growth.
- PC6.** To visit field experiments to evaluate the effectiveness of bioremediation techniques for degrading specific pollutants in soil.
- PC7.** To assess and evaluate the indicators (Bioconcentration factor and Translocation factor) for effective Phytoremediation strategies.
- PC8.** Patent Search and Analysis: Conduct patent searches using online databases and analyze patent documents to assess the novelty and patentability of inventions.
- PC9.** Case Studies and Simulations: Analyze real-world case studies involving IPR disputes and participate in simulations to develop negotiation and decision-making skills.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles of bioremediation using plants and microorganisms.
- KU2.** Phytoremediation techniques: phytoextraction, phytostabilization, phytodegradation, rhizofiltration.

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- KU3.** Role of salt-tolerant species in saline soil remediation.
- KU4.** Microbial functions and factors affecting pollutant degradation.
- KU5.** Basics and types of IPR for innovation protection.
- KU6.** Patent search, analysis, and IPR case study evaluation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply bioremediation techniques using plants and microbes in field and lab settings.
- GS2.** Evaluate phytoremediation effectiveness using BCF and TF indicators.
- GS3.** Conduct field experiments to assess soil pollutant degradation.
- GS4.** Perform patent searches and analyze documents for novelty and patentability.
- GS5.** Interpret IPR concepts and apply them to protect scientific innovations.
- GS6.** Analyze case studies and engage in simulations to enhance decision-making and negotiation skills.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Soil Pollution: Mitigation and Technologies</i>	60	40	-	-
PC1. Understand the principles and mechanisms of bioremediation, including the use of microorganisms and plants to degrade or remove pollutants from soil.	10	-	-	-
PC2. Explore the use of plants for the remediation of contaminated soil, including phytoextraction, Phyto stabilization, phytodegradation, and rhizofiltration.	10	-	-	-
PC3. Learn about the use of salt-tolerant plants and microorganisms to improve soil quality and productivity in saline environments.	10	-	-	-
PC4. Understand the role of microorganisms in bioremediation and the factors influencing their effectiveness in degrading various pollutants.	20	-	-	-
PC5. Understand the concept and types of IPR, including patents, trademarks, copyrights, and trade secrets, and their importance in protecting innovations and promoting economic growth.	10	-	-	-
PC6. To visit field experiments to evaluate the effectiveness of bioremediation techniques for degrading specific pollutants in soil.	-	10	-	-
PC7. To assess and evaluate the indicators (Bioconcentration factor and Translocation factor) for effective Phytoremediation strategies.	-	10	-	-
PC8. Patent Search and Analysis: Conduct patent searches using online databases and analyze patent documents to assess the novelty and patentability of inventions.	-	10	-	-
PC9. Case Studies and Simulations: Analyze real-world case studies involving IPR disputes and participate in simulations to develop negotiation and decision-making skills.	-	10	-	-
NOS Total	60	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N6905
NOS Name	Prepare Strategies for Soil Pollution Mitigation and Adoption of Advanced Technologies
Sector	Green Jobs
Sub-Sector	
Occupation	Soil Scientist
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N6901.Introduce Soil Composition, Quality, and Health Implications	64	36	-	-	100	20
SGJ/N6902.Assess and Classify Soil Properties	45	55	-	-	100	20
SGJ/N6903.Identify techniques for Soil Sampling and Ensuring Quality Standards	58	42	-	-	100	20
SGJ/N6904.Apply Statistical and GIS Tools in Soil Analysis	60	40	-	-	100	20
SGJ/N6905.Prepare Strategies for Soil Pollution Mitigation and Adoption of Advanced Technologies	60	40	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	307	243	-	-	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.